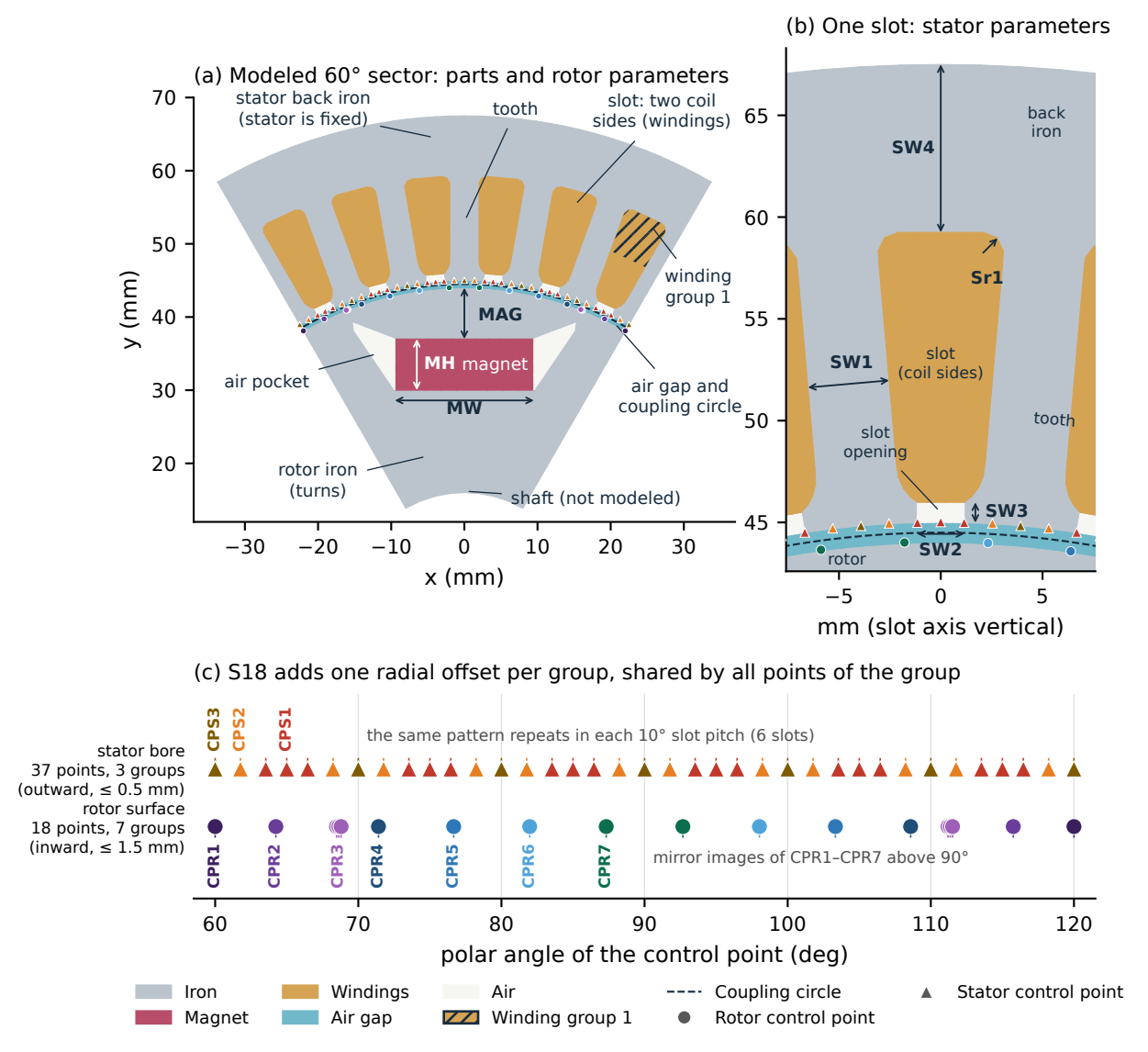

(a) Modeled 60° sector: parts and rotor parameters
stator back iron
(stator is fixed)
tooth
slot: two coil
sides (windings)
winding
group 1
MAG
MH magnet
MW
air pocket
air gap and
coupling circle
rotor iron
(turns)
shaft (not modeled)
y (mm)
x (mm)
(b) One slot: stator parameters
back
iron
SW4
Sr1
SW1
slot
(coil sides)
tooth
slot
opening
SW3
SW2
rotor
mm (slot axis vertical)
(c) S18 adds one radial offset per group, shared by all points of the group
stator bore
37 points, 3 groups
(outward, ≤ 0.5 mm)
rotor surface
18 points, 7 groups
(inward, ≤ 1.5 mm)
CPS3
CPS2
CPS1
the same pattern repeats in each 10° slot pitch (6 slots)
CPR1
CPR2
CPR3
CPR4
CPR5
CPR6
CPR7
mirror images of CPR1–CPR7 above 90°
polar angle of the control point (deg)
Iron
Magnet
Windings
Air gap
Air
Winding group 1
Coupling circle
Rotor control point
Stator control point